(Table 1, contd)

System or Component Name	Limiting Condition	Criteria for Action	Required Action
Braking System	Inadequate braking performance	System does not pass Braking System - Test(s) included in Maintenance Section or in the Testing and Adjusting Manual	Contact your Cat ® dealer to inspect and, if necessary, repair the brake system.
Cooling System	The coolant temperature is too high.	Monitoring System displays Warning Category 3	Stop the engine immediately. Check the coolant level and check the radiator for debris. Refer to Operation and Maintenance Manual, Cooling System Coolant Level - Check. Check the fan drive belts for the water pump. Refer to Operation and Maintenance Manual, Belts - Inspect/Adjust/ Replace. Make any necessary repairs.
Engine Oil System	A problem has been detected with the engine oil pressure.	Monitoring System displays Warning Category 3	If the warning stays on during low idle, stop the engine and check the engine oil level. Perform any necessary repairs as soon as possible.
Engine system	An engine fault has been detected by the engine ECM.	Monitoring System displays Warning Category 3	Stop the engine immediately. Contact your Cat ® dealer for service.
Fuel System	A problem has been detected with the fuel system.	Monitoring System displays Warning Category 3	Stop the engine. Determine the cause of the fault and perform any necessary repairs.
Hydraulic Oil System	The hydraulic oil temperature is too high.	Monitoring System displays Warning Category 3	Stop the engine immediately. Check the hydraulic oil level and check the hydraulic oil cooler for debris. Perform any necessary repairs as soon as possible.
Steering System	A problem has been detected with the steering system. (If equipped with steering system monitoring.)	Monitoring System displays Warning Category 3	Move machine to a safe location and stop the engine immediately. Contact your Cat ® dealer to inspect and, if necessary, repair the steering system.
Overall Machine	Machine service is required.	Monitoring System displays Warning Category 3	Stop the engine immediately. Contact your Cat ® dealer for service.

Machine Operation

For machines that are equipped with a ROPS. A seat belt should be worn always during machine operation to prevent serious injury or death in the event of an accident or machine overturn. Failure to wear a seat belt during machine operation may result in serious injury or death.

For machines that are NOT equipped with a ROPS, falling from a machine could cause serious injury or death. Seat belts should be worn when driving the machine on public roads.

Only operate the machine while you are in a seat. Only operate the controls while the engine is running.

While you operate the machine slowly in an open area, check for proper operation of all controls and all protective devices.

Make sure that no personnel will be endangered before you move the machine.

Do not allow riders on the machine unless the machine has the following equipment:

- additional seat
- additional seat belt

Never use the work tool as a work platform.

Note any needed repairs during machine operation. Report any needed repairs.

Carry attachments approximately 40 cm (15 inches) above ground level. Do not go close to the edge of a cliff, an excavation, or an overhang.

If the machine begins to sideslip downward on a grade, immediately remove the load and turn the machine downhill.

Avoid any conditions that can lead to tipping the machine. The machine can tip when you work on hills, on banks, and on slopes. Also, the machine can tip when you cross ditches, ridges, or other unexpected obstructions.

Avoid operating the machine across the slope. When possible, operate the machine up the slopes and down the slopes.

Maintain control of the machine. Do not overload the machine beyond the machine capacity.

Never straddle a wire cable. Never allow other personnel to straddle a wire cable.

Know the maximum dimensions of your machine.

i06299648

Engine Stopping

SMCS Code: 1000; 7000

Do not stop the engine immediately after the machine has been operated under load. Stopping the engine immediately can cause overheating and accelerated wear of engine components.

After the machine is parked and the parking brake is engaged, allow the engine to run at low idle for 5 minutes before shutdown. Running the engine allows hot areas of the engine to cool gradually.

i04725627

Parking

SMCS Code: 7000

Park on a level surface. If you must park on a grade, chock the wheels.

Move the propel control lever to the NEUTRAL position.

Lower all attachments to the ground.

Engage the parking brake.

Stop the engine.

Turn the engine start switch to the OFF position and remove the engine start switch key.

Always turn the battery disconnect switch to the OFF position before leaving the machine.

If the machine will not be operated for a week or more, remove the battery disconnect switch key.

i03647047

Slope Operation

SMCS Code: 7000

Machines that are operating safely in various applications depend on these criteria: the machine model, configuration, machine maintenance, operating speed of the machine, conditions of the terrain, fluid levels and tire inflation pressures. The most important criteria are the skill and judgment of the operator.

A well trained operator that follows the instructions in the Operation and Maintenance Manual has the greatest impact on stability. Operator training provides a person with the following abilities: observation of working and environmental conditions, feel for the machine, identification of potential hazards and operating the machine safely by making appropriate decisions.

When you work on side hills and when you work on slopes, consider the following important points:

Speed of travel – At higher speeds, forces of inertia tend to make the machine less stable.

Roughness of terrain or surface – The machine may be less stable with uneven terrain.

Direction of travel – Avoid operating the machine across the slope. When possible, operate the machine up the slopes and operate the machine down the slopes. In order to achieve the best compaction performance and steering control on a slope, operate a vibratory soil compactor with the drum end of the machine downhill.

Compacting with drum beyond edge of surface – Machines with solid drums can tip suddenly as the center of balance of the machine moves beyond the edge of the compacted surface. Slow down and pay close attention when operating with the drum extended beyond the edge of the compacted surface. Minimize the amount of the drum that extends beyond the edge.

Over compacting – When the material is fully compacted, and the vibratory system is activated, the drum may bounce on the compacted surface. If the machine is on an incline, this can cause the machine to move down the slope with the force of gravity. Reduce the vibratory amplitude or shut off vibration if the drum is bouncing on the compacted surface.

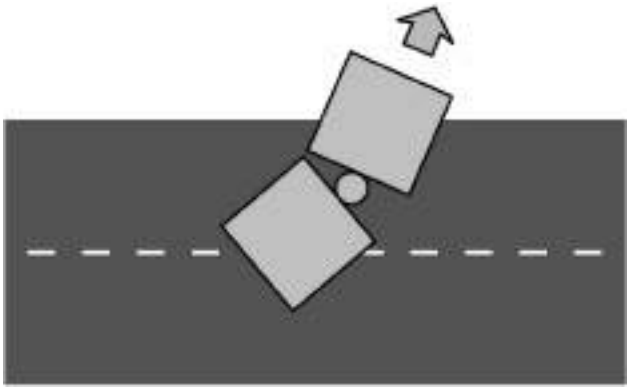


Illustration 27 g01956702

The appropriate method for driving off a compacted surface

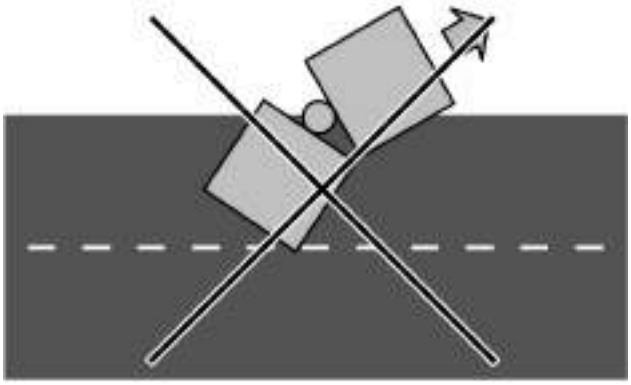


Illustration 28 g01957405

The improper method for driving off a compacted surface

Steering angle – The steering angle affects the lateral balance of an articulated machine. When the machine is driven off the compacted mat, always exit by turning the machine toward the edge. Refer to illustration 27 . Do not turn away from the edge when the machine is driven off the compacted mat. Refer to illustration 28 .

Mounted equipment – Balance of the machine may be impeded by the following components: equipment that is mounted on the machine, machine configuration, weights and counterweights.

Nature of surface – Ground that has been newly filled with earth may collapse from the weight of the machine. The vibratory action of a vibratory compactor can increase the tendency of material on the edge of an incline to collapse.

Surface material – Rocks and moisture of the surface material may drastically affect the machine's traction and machine's stability. Rocky surfaces may promote side slipping of the machine.

Operated equipment – Be aware of performance features of the equipment in operation and the effects on machine stability.

Operating techniques – Keep all attachments low to the ground for optimum stability.

Machine systems have limitations on slopes – Slopes can affect the proper function and operation of the various machine systems. These machine systems are needed for machine control.

Note: Safe operation on steep slopes may require special machine maintenance. Excellent skill of the operator and proper equipment for specific applications are also required. Consult the Operation and Maintenance Manual for the proper fluid level requirements and intended use for the machine.

i08229294

Equipment Lowering with Engine Stopped

SMCS Code: 7000-II; 7000

Before lowering any equipment with the engine stopped, clear the area around the equipment of all personnel. The procedure to use will vary with the type of equipment to be lowered. Keep in mind most systems use a high pressure fluid or air to raise or lower equipment. The procedure will cause high pressure air, hydraulic, or some other media to be released in order to lower the equipment. Wear appropriate personal protective equipment and follow the established procedure in the Operation and Maintenance Manual, “Equipment Lowering with Engine Stopped” in the Operation Section of the manual.

i08684845

Sound Information and Vibration Information

SMCS Code: 7000

Sound Level Information

Hearing protection may be needed when the machine is operated with an open operator station, in a noisy environment, with a cab that is not properly maintained, or when the doors and windows are open for extended periods of time.

Table 2

Sound Level		Test Method
Operator Sound Pressure Level	88 dB(A)	“ISO 6396:2008” (1)
Exterior Sound Power Level	104 dB (A)	“ISO 6395:2008” (2)

(1) The measurement was conducted at 70% of the maximum engine cooling fan speed. The sound level may vary at different engine cooling fan speeds. The measurement was conducted with the cab doors and the cab windows closed. The cab was properly installed and maintained.

(2) The measurement was conducted at 70% of the maximum engine cooling fan speed. The sound level may vary at different engine cooling fan speeds.

The sound levels listed above include both measurement uncertainty and uncertainty due to production variation.

Sound Level Information for Machines Required by the Applicable Regional Regulations

- **European Union Countries**

- **United Kingdom**
- **Eurasian Economic Union Countries**
- **Ukraine**
- **Countries that Adopt the “EU Directives”**

The information below applies to only the machine configurations that contain regional product marking on or near the Product Identification Plate noted in the “Regional Product Marking” section of this manual.

Table 3

Declared Dynamic Operator Sound Pressure Level		
Region	**Sound Level**	**Test Method**
European Union	88 dB(A)	“ISO 6396:2008” (1)
United Kingdom	88 dB(A)	“ISO 6396:2008” (1)
Eurasian Economic Union	88 dB(A)	“ISO 6396:2008” (1)

(1) The measurement was conducted at 70% of the maximum engine cooling fan speed. The sound level may vary at different engine cooling fan speeds. The measurement was conducted with the cab doors and the cab windows closed. The cab was properly installed and maintained.

Table 4

Declared Exterior Sound Power Level		
Region	**Sound Level**	**Test Method**
European Union	104 dB(A)	“ISO 6395:1988” (1)
United Kingdom	104 dB(A)	“ISO 6395:1988” (1)
Eurasian Economic Union	104 dB(A)	“ISO 6395:2008” (1)
Ukraine	104 dB(A)	“ISO 6395:1988” (1)

(1) The measurement was conducted at 70% of the maximum engine cooling fan speed. The sound level may vary at different engine cooling fan speeds.

The declared sound levels listed above include both measurement uncertainty and uncertainty due to production variation.

The machine sound power level meets the criteria that are specified in the applicable regional regulation. For example:

- “European Directive 2000/14 EC” amended by “2005/88/EC”
- “United Kingdom 2001 No. 1701” amended by “2005 No. 3525”
- “Ukraine Technical Regulation of the Noise Emission in the Environment by Equipment for Use Outdoors”

The criteria are specified on the certificate of the conformance and the accompanying labels.

Vibration Information Applicable to Regional Regulations

- “**European Union Directive: 2002/44/EC - Physical Agents (Vibration)** ”
- “**United Kingdom: 2005 No. 1093 - The Control of Vibration at Work Regulation 2005** ”

Vibration Data for Asphalt Compactors

Information Concerning Hand/Arm Vibration Level

When the machine is operated according to the intended use, the hand/arm vibration of this machine is below 2.5 meter per second squared.

Information Concerning Whole Body Vibration Level

This section provides vibration data and a method for estimating the vibration level for Asphalt Compactors.

Note: Vibration levels are influenced by many different parameters. Many items are listed below.

- Operator training, behavior, mode, and stress
- Job site organization, preparation, environment, weather, and material
- Machine type, quality of the seat, quality of the suspension system, attachments, and condition of the equipment

It is not possible to get precise vibration levels for this machine. The expected vibration levels can be estimated with the information in the table below to calculate the daily vibration exposure. A simple evaluation of the machine application can be used.

Estimate the vibration levels for the three vibration directions. For typical operating conditions, use the average vibration levels as the estimated level. With an experienced operator and smooth terrain, subtract the Scenario Factors from the average vibration level to obtain the estimated vibration level. For aggressive operations and severe terrain, add the Scenario Factors to the average vibration level to obtain the estimated vibration level.

Note: All vibration levels are in meter per second squared.

Table 5

"ISO/TR 25398 reference table - Equivalent vibration levels of whole body vibration emission for earthmoving equipment."							
Machine Type	**Typical Operating Activity**	**Vibration Levels**			**Scenario Factors**		
		X axis	**Y axis**	**Z axis**	**X axis**	**Y axis**	**Z axis**
Vibratory Asphalt Compactor	vibration ON	"0, 33"	"0, 40"	"0, 48"	"0, 11"	"0, 08"	"0, 14"
	vibration OFF	"0, 35"	"0, 43"	"0, 36"	"0, 13"	"0, 20"	"0, 19"

Note: Refer to "ISO/TR 25398 Mechanical Vibration - Guideline for the assessment of exposure to whole body vibration of ride on operated earthmoving machines" for more information about vibration. This publication uses data that is measured by international institutes, organizations, and manufacturers. This document provides information about the whole body exposure of operators of earthmoving equipment. Refer to Operation and Maintenance Manual, SEBU8257. Consult regional or local legislation for more information about machine vibration levels.

The Cat ® suspension seat meets the criteria of "ISO 7096". This represents vertical vibration level under severe operating conditions. This seat is tested with the input "spectral class EM1". The seat has a transmissibility factor of "SEAT<1.1".

The whole body vibration level of the machine varies. There is a range of values. The low value is 0.5 meter per second squared. The machine meets the short-term level for the design of the seat in "ISO 7096". The value is 1.13 meter per second squared for this machine.

Guidelines for Reducing Vibration Levels on Earthmoving Equipment

Properly adjust machines. Properly maintain machines. Operate machines smoothly. Maintain the conditions of the terrain. The following guidelines can help reduce the whole body vibration level:

1. Use the right type and size of machine, equipment, and attachments.

2. Maintain machines according to the manufacturer recommendations.

a. Tire pressures

b. Brake and steering systems

c. Controls, hydraulic system, and linkages

d. Suspension components

3. Keep the terrain in good condition.

a. Remove any large rocks or obstacles.

b. Fill any ditches and holes.

c. Provide machines and schedule time to maintain the conditions of the terrain.

4. Use a seat that meets "ISO 7096". Keep the seat maintained and properly adjusted.

a. Adjust the seat and suspension for the weight and the size of the operator.

b. Inspect and maintain the seat suspension and adjustment mechanisms.

5. Perform the following operations smoothly.

a. Steer

b. Brake

c. Accelerate

d. Shift the gears

6. Move the attachments smoothly.

7. Adjust the machine speed and the route to minimize the vibration level.

a. Drive around obstacles and rough terrain.

b. Slow down when necessary, to go over rough terrain.

8. Minimize vibrations for a long work cycle or a long travel distance.

a. Use machines that are equipped with suspension systems.

b. Use the ride control system when available.

c. If no ride control system is available, reduce speed to prevent bounce.

d. Haul the machines between workplaces.

9. Less operator comfort may be caused by other risk factors. The following guidelines can be effective to provide better operator comfort:

a. Adjust the seat and adjust the controls to achieve good posture.

b. Adjust the mirrors to minimize twisted posture.

c. Provide breaks to reduce long periods of sitting.

d. Avoid jumping from the cab.

e. Minimize repeated handling of loads and lifting of loads.

f. Minimize any shocks and impacts during sports and leisure activities.

Sources

The vibration information and calculation procedure are based on “ISO/TR 25398 Mechanical Vibration - Guideline for the assessment of exposure to whole body vibration of ride on operated earthmoving machines”. Harmonized data is measured by international institutes, organizations, and manufacturers.

This literature provides information about assessing the whole body vibration exposure of operators of earthmoving equipment. The method is based on measured vibration emission under real working conditions for all machines.

Check the original directive. This document summarizes part of the content of the applicable law. This document is not meant to substitute the original sources. Other parts of these documents are based on information from the United Kingdom Health and Safety Executive.

Refer to Operation and Maintenance Manual, SEBU8257. Consult regional or local legislation for more information about vibration.

Consult your local Cat dealer for more information about machine features that minimize vibration levels. Consult your local Cat dealer about safe machine operation.

Use the following web site to find your local dealer:

Caterpillar, Inc.
www.cat.com

Product Information Section

General Information

i08518284

Specifications

SMCS Code: 7000

Intended Use

This roller is a self-propelled vibratory compactor. This roller consists of one or more metallic cylindrical bodies (drums) that are used for compaction. This roller may also consist of rubber tires that are used for compaction. This roller is used to compact materials such as crushed rock, earth, asphalt, or gravel through rolling and/or vibrating action of the roller.

Expected Life

The expected life, defined as total machine hours, of this machine is dependent upon many factors. The factors include the desire of the owner to rebuild the machine back to factory specifications. The expected life interval of this machine is 8,000 service hours The expected life interval corresponds to the service hours to engine overhaul or replacement. Service hours to engine overhaul or replacement may vary based on overall machine duty cycle. At the expected life interval, remove the machine from operation. Consult your Cat® dealer for inspect, repair, rebuild, install remanufactured, install new components, or disposal options and to establish a new expected life interval. If a decision is made to remove this machine from service, refer to “Decommissioning and Disposal“ for more information.

The following items are required to obtain an economical expected life of this machine:

- Perform regular preventive maintenance procedures as described in the Operation and Maintenance Manual.
- Perform machine inspections as described in the Operation and Maintenance Manual and correct any problems discovered.
- Perform system testing as described in the Operation and Maintenance Manual and correct any problems discovered.
- Ensure that all service letters are addressed in the time intervals described in the letters.
- Ensure that machine application conditions comply with Caterpillar’s recommendations.
- Ensure that the operating weight does not exceed limits set by manufacturer.
- Ensure that all frame cracks are identified, inspected, and repaired to prevent further development.

Dimensions

Table 6

CB2.5 GC Vibratory Compactor	
Maximum Operating Weight	3021 kg (6660 lb)
Maximum Length of Machine	2673 mm (105 inch)
Width of the Machine	1112 mm (44 inch)
Height of the Machine with canopy and beacon	2869 mm (113 inch)
Width of Drum	1000 mm (40 inch)

Table 7

CB2.7 GC Vibratory Compactor	
Maximum Operating Weight	3198 kg (7050 lb)
Maximum Length of Machine	2673 mm (105 inch)
Width of the Machine	1312 mm (52 inch)
Height of the Machine with canopy and beacon	2869 mm (113 inch)
Width of Drum	1200 mm (47 inch)

Table 8

CC2.7 GC Vibratory Compactor	
Maximum Operating Weight	2779 kg (6127 lb)
Maximum Length of Machine	2673 mm (105 inch)
Width of the Machine	1312 mm (52 inch)
Height of the Machine with canopy and beacon	2869 mm (113 inch)
Width of Drum	1200 mm (47 inch)

Note: The following items are included in the operating machine weight: lubricants, coolant, half full fuel tank, full hydraulic tank, half full water tank, and an 80 kg (176 lb) operator.

Identification Information

i08684866

Plate Locations and Film Locations

SMCS Code: 1000; 7000

The Product Information Number (PIN) will be used to identify a powered machine that is designed for an operator to ride.

Caterpillar products such as engines, transmissions, and major attachments that are not designed for an operator to ride are identified by Serial Numbers.

The certification plate (CE) is used to verify that the product conforms to all the requirements that were established by a country or a group of countries. The product is tested by a certified testing group to verify conformance.

For quick reference, record the identification numbers in the spaces that are provided below the illustration.

Product Identification Number (PIN)

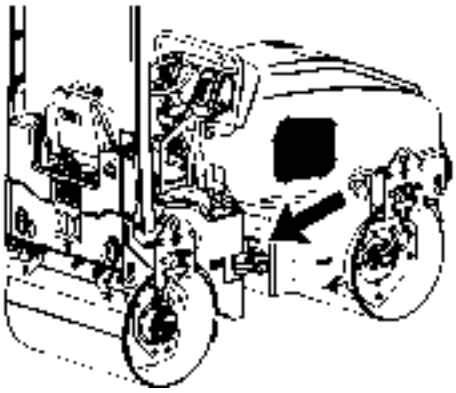


Illustration 29 g03162376

The Product Identification Number (PIN) plate is at the rear of the machine on the left side. The plate may have the following information:

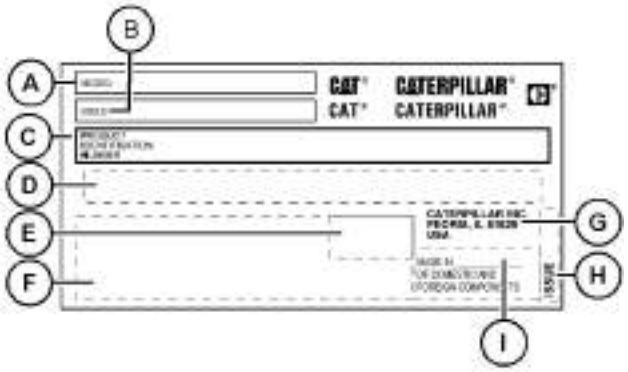


Illustration 30 g06201159

Manufacturer Name and Address _____

Model (A)_____

Build (B)_____

Product Identification Number (C)_____

Bar Code (D)_____

Month and/or Year of Manufacture Plate (If Required) (E)_____

Regional Certification Plate (If Required) (F)_____

Address of Manufacturer (G)_____

Issue (H)_____

Country of Origin Info Plate (If Required) (I)_____

Local regulation may require documentation of the month and/or year of manufacture in the Operation and Maintenance Manual. Comply with these regulations.

Regional Product Marking (If Equipped)



Illustration 31 g06650998

Regional marking plate

This plate is positioned on the bottom-left side of the PIN plate or near the PIN plate.

Note: The regional marking plate or plates are installed on machines that meet the applicable requirements that were effective at that time and may differ from the one shown above.

Regional product marking may include one or more of the following:

CE mark

UKCA mark

EAC mark

Gulf Standardization Organization (GSO) mark

Ukraine mark

The following information may be stamped onto the regional product marking plate. For quick reference, record this information in the spaces that are provided below:

- Engine Power Primary Engine (kW)_____
- Engine Power for Additional Engine (If Equipped)_____
- Typical Machine Operating Weight (kg)_____
- Month and/or Year of Manufacture_____
- Machine Type_____

Eurasian Economic Union

Manufacturer Information

Manufacturer:

Caterpillar Inc.,
9401 85th Avenue North
Brooklyn park MN, 55445-2199

Entity authorized by the manufacturer at the territory of Eurasian Economic Union:

Caterpillar Eurasia LLC
75, Sadovnicheskaya Emb.
Moscow 115035, Russia

Sound Certification

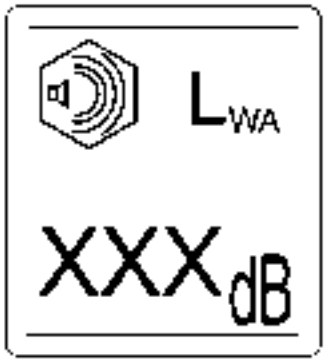


Illustration 32 g06675270

Sound certification film

A typical example of this film is shown.

A certification film is used to verify the environmental sound certification on machines that are certified to the regional requirements. A film installed on your machine will have a value. The value that is listed on the film indicates the guaranteed exterior sound power level (L_{wa}) at the time of manufacture for the conditions that are specified in the following sound test procedures:

- “ISO 6395:1988”
- “European Directive 2000/14 EC” amended by “2005/88/EC”
- “United Kingdom 2001 No. 1701” amended by “2005 No. 3525”

i08756279

Emissions Certification Film

SMCS Code: 1000; 7000; 7405

Consult your Cat dealer for an Emission Control Warranty Statement.

The emission certification film is located on the engine or inside the engine enclosure.

Declaration of Conformity
(European Union)

SMCS Code: 1000; 7000

Table 9

An EU Declaration of Conformity document was provided with the machine if the machine was manufactured to comply with specific requirements for the European Union. To determine the details of the applicable Directives, review the complete EU Declaration of Conformity provided with the machine. The extract shown below from an EU Declaration of Conformity for machines that are declared compliant to “2006/42/EC” applies only to those machines originally “CE” marked by the manufacturer listed and which have not since been modified.

ORIGINAL EU DECLARATION OF CONFORMITY

Manufacturer: CATERPILLAR PAVING PRODUCTS INC. 9401 85th Ave. North Brooklyn Park, MN 55445 USA

Person authorized to compile the Technical File and to communicate relevant part (s) of the Technical File to the Authorities of European Union Member States on request:

Standards & Regulations Manager,Caterpillar France S.A.S,,
40 Avenue Leon-Blum 38000 Grenoble, France

I, the undersigned, _____, hereby certify that the construction equipment specified hereunder

Description:	Generic Denomination:	Paving Equipment
	Function:	Vibratory Roller
	Model/Type:	CB2.5 GC, CB2.7 GC, CC2.7 GC
	Serial Number:	
	Commercial Name:	Caterpillar

Fulfills all the relevant provisions of the following Directives

Directives	Notified Body	Document No.
2006/42/EC	N/A	
2000/14/EC amended by 2005/88/EC, Note (1)		
2014/30/EC	N/A	

Note (1) Annex -_____ Guaranteed Sound Power Level -_____dB (A)
Representative Equipment Type Sound Power Level - _____dB (A)
Engine Power per ____-____ kW Rated engine speed - _____ rpm
Technical Documentation accessible through person listed above authorized to compile the Technical File

Done at: **Signature**

Date: **Name/Position**

Note: The above information was correct as of August 2022, but may be subject to change, refer to the individual declaration of conformity issued with the machine for exact details.

Declaration of Conformity
(Great Britain)

SMCS Code: 1000; 7000

Table 10

A Declaration of Conformity document was provided with the machine if the machine was manufactured to comply with specific requirements for the Great Britain. To determine the details of the applicable legislation, review the complete Declaration of Conformity provided with the machine. The extract shown below from a Great Britain Declaration of Conformity for machines that are declared compliant to 2008 No. 1597 applies only to those machines originally “UKCA” marked by the manufacturer listed and which have not since been modified.

DECLARATION OF CONFORMITY

Manufacturer: CATERPILLAR PAVING PRODUCTS INC. 9401 85th Ave. North Brooklyn Park, MN 55445 USA

Person authorized to compile the Technical File and to communicate relevant part (s) of the Technical File to the Authorities on request:

Standards & Regulations Manager Caterpillar France SAS
40 Avenue Leon-Blum 38000 Grenoble, France

I, the undersigned, _____, hereby certify that the construction equipment specified hereunder

Description:	Generic Denomination:	Paving Equipment
	Function:	Vibratory Roller
	Model/Type:	CB2.5 GC, CB2.7 GC, CC2.7 GC
	Serial Number:	
	Commercial Name:	Caterpillar

Fulfills all the relevant provisions of these regulations and/or other enactments as listed below:

Legislation	Approved Body	Document No.
2008 No. 1597	N/A	
2016 No. 1091	N/A	
2001 No. 1701 amended by 2005 No. 3525, Note (1)	Note (2)	

Note (1) Schedule -_____ Guaranteed Sound Power Level -_____dB (A)
Representative Equipment Type Sound Power Level - _____dB (A)
Engine Power per ____-____ kW Rated engine speed - _____ rpm
Technical Documentation accessible through person listed above authorized to compile the Technical File

Note (2) If applicable, information related to Approved Body.

Done at: **Signature**

Date: **Name/Position**

Note: The above information was correct as of August 2022, but may be subject to change, refer to the individual declaration of conformity issued with the machine for exact details.

Operation Section

Before Operation

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Mounting and Dismounting

SMCS Code: 7000

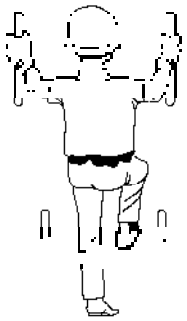


Illustration 33 g00037860
Typical example

Mount the machine and dismount the machine only at locations that have steps and/or handholds. Before you mount the machine, clean the steps and the handholds. Inspect the steps and handholds. Make all necessary repairs.

Face the machine whenever you get on or off the machine.

Maintain a three-point contact with the steps and with the handholds.

Note: Three-point contact can be two feet and one hand. Three-point contact can also be one foot and two hands.

Do not mount a moving machine. Do not dismount a moving machine. Never jump off the machine. Do not carry tools or supplies when you try to mount the machine or when you try to dismount the machine. Use a hand line to pull equipment onto the platform. Do not use any controls as handholds when you enter the operator compartment or when you exit the operator compartment.

Machine Access System Specifications

The machine access system has been designed to meet the intent of the technical requirements in “ISO 2867:2011 Earth-moving Machinery – Access Systems”. The access system provides for operator access to the operator station and to conduct the maintenance procedures described in Maintenance section.

Alternate Exit

Machines that are equipped with cabs have alternate exits. For additional information, see Operation and Maintenance Manual, “Alternate Exit”.

i08466304

Daily Inspection

SMCS Code: 1000; 7000

For a maximum service life of the machine, complete a thorough walk-around inspection before you mount the machine and before you start the engine.

Inspect the area around the machine and under the machine. Look for loose bolts, trash buildup, oil, coolant leakage, broken parts, or worn parts.

Note: Watch closely for leaks. If you observe a leak, find the source of the leak and correct the leak. If you suspect a leak or you observe a leak, check the fluid levels more frequently.

Inspect the condition of the equipment and of the hydraulic components.

Check the condition of the tires. Adjust the inflation pressure, if necessary.

Check all the oil levels, all the coolant levels, and all the fuel levels.

Remove any trash buildup and debris. Make all necessary repairs before you operate the machine.

Make sure that all covers and guards are securely attached.

Adjust the mirrors for the correct rear view of the machine.

Grease all the fittings that need to be serviced daily.

Daily, perform the procedures that are applicable to your machine:

- Backup Alarm - Test
- Beacon and Lights - Inspect
- Cooling System Level - Check
- Drum Scrapers - Inspect/Adjust/Replace

- Engine Oil Level - Check
- Hydraulic System Oil Level - Check
- Indicators and Gauges - Test
- Neutral Start Switch - Test
- Seat Belt - Inspect
- Water Spray System Filter - Clean

i08579990

Steering Frame Lock

SMCS Code: 7506

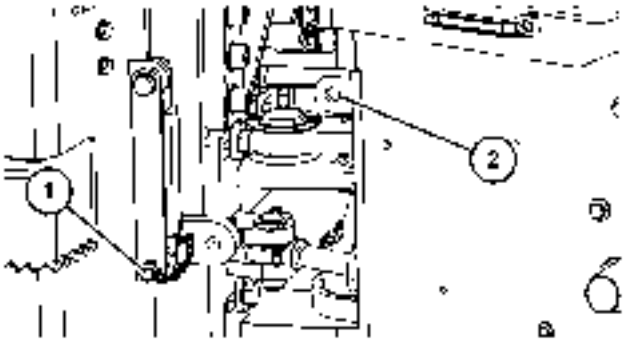


Illustration 34 g06183886

Unlocked position

To lock the frame, remove pin (1) and rotate the bracket up to align with eye hole (2). Insert pin (1) into eye hole (2) and lock it in place using the clip.

Machine Operation

i08466539

Seat

SMCS Code: 7312; 7324

Note: Adjust the seat for a new operator or at the beginning of each shift.

The operator should sit in the seat to determine the correct positioning. Adjust the seat so that the operator is allowed full travel of all the controls.

Lock the seat into position before you operate the machine. Locking the seat will prevent unexpected movement of the seat.

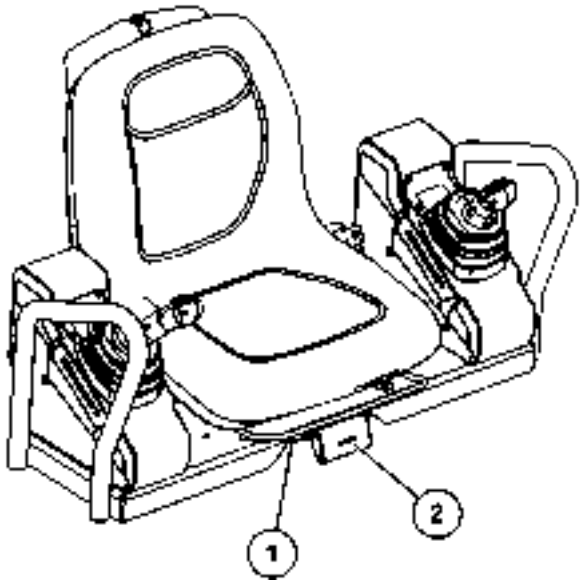


Illustration 35 g06673638

Standard seat

(1) Forward/backward slide adjustment
(2) Left/Right slide adjustment

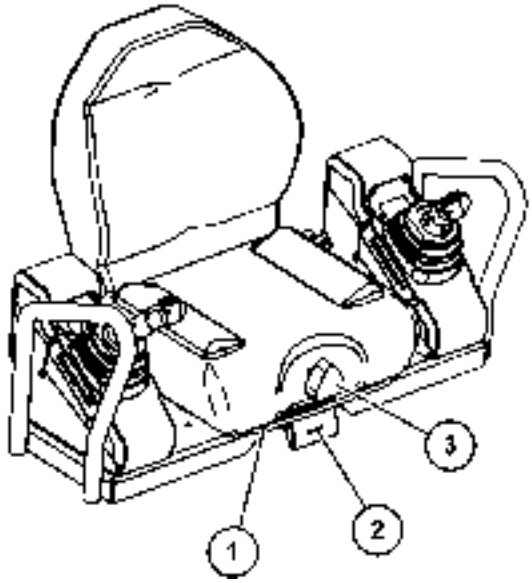


Illustration 36 g06673645

Optional seat (If Equipped)

(1) Forward/backward slide adjustment
(2) Left/Right slide adjustment
(3) Weight adjustment (if equipped)

Forward/Back Slide Adjustment (1) – Lift the bar at the front of the seat. Slide the seat until the desired position is attained. Release the bar to lock the seat into position.

Left/Right Slide Adjustment (2) – Lift the handle at the front of the seat. Slide the seat until the desired position is attained. Release the handle to lock the seat into position.

Weight Adjustment (3) (If Equipped) – Rotate the dial to increase or decrease the seat stiffness.

Operator Presence (If Equipped)

The operator presence detects when the operator is present in the operator seat.

If the operator is not present in the operator seat.

1. The level 1 event will occur. A solid audible alarm will be heard from the machine.
2. Then Level 2 event will occur. The vibe system will be switched to OFF position. The engine will be set to low idle and the operator out of seat indicator will get energized in the display event.

Note: To re-enable propel. The park brake switch must be moved to park brake apply position and then moved to park brake release position. The propel lever should be in NEUTRAL position. Refer to “Operator Controls“ for more information.

Note: To re-enable the vibe system. The vibe system switch must be pressed again. Refer to “Operator Controls“ for more information.

i04575638

Seat Belt

SMCS Code: 7327

Note: This machine was equipped with a seat belt when the machine was shipped from Caterpillar. At the time of installation, the seat belt and the instructions for installation of the seat belt meet the SAE J386 and ISO 6683 standards. Consult your Cat dealer for all replacement parts.

Always check the condition of the seat belt and the condition of the mounting hardware before you operate the machine.

Seat Belt Adjustment for Retractable Seat Belts

Fastening The Seat Belt

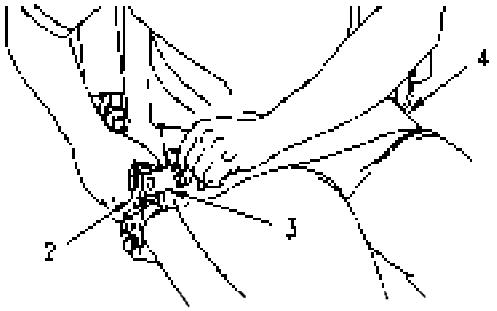


Illustration 37 g00867598

Pull seat belt (4) out of the retractor in a continuous motion.

Fasten seat belt catch (3) into buckle (2). Make sure that the seat belt is placed low across the lap of the operator.

The retractor will adjust the belt length and the retractor will lock in place. The comfort ride sleeve will allow the operator to have limited movement.

Releasing The Seat Belt

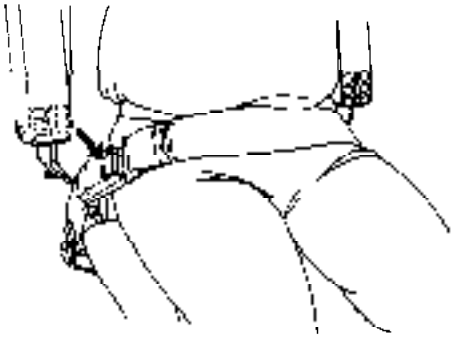


Illustration 38 g00039113

Push the release button on the buckle in order to release the seat belt. The seat belt will automatically retract into the retractor.

Extension of the Seat Belt

⚠ WARNING

When using retractable seat belts, do not use seat belt extensions, or personal injury or death can result.

The retractor system may or may not lock up depending on the length of the extension and the size of the person. If the retractor does not lock up, the seat belt will not retain the person.

Longer, non-retractable seat belts and extensions for the non-retractable seat belts are available.

Caterpillar requires only non-retractable seat belts to be used with a seat belt extension.

Consult your Cat dealer for longer seat belts and for information on extending the seat belts.

i08518330

Operator Controls

SMCS Code: 7000; 7300; 7301; 7451

Note: Your machine may not be equipped with all the controls that are discussed in this topic.

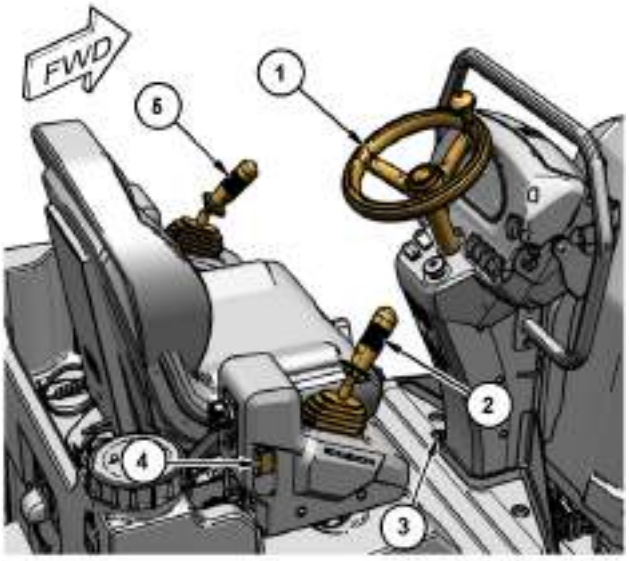


Illustration 39 g06673749

(1) Steering wheel
(2) Right propel control
(3) Traction Control Switch (if equipped)
(4) 12V power receptacle
(5) Left propel control (if equipped)



Illustration 40 g06610378

(6) Emergency stop knob
(7) Work lights
(8) Front/Position lights
(9) Turn signal
(10) Hazard lights
(11) Gauges and indicators
(12) Parking brake
(13) Horn
(14) Water spray system
(15) Engine start switch
(16) Vibratory mode / Emulsion switch
(17) Engine speed
(18) Beacon
(19) Digital display

Steering Wheel (1)

Steering wheel (1) controls the directional steering of the machine. The machine will turn in the same direction as steering wheel (1).

Note: The steering frame lock must be unlocked to steer the machine.

Right Propel Control (2)

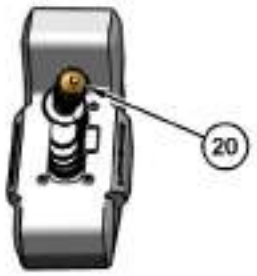


Illustration 41 g06611008

(20) On/off switch for the vibratory system

On/Off Switch for the vibratory system (20) – To turn on the vibration system, press the switch. to stop the vibration system, press the switch again.

Note: This machine is set for auto vibe from the factory. Therefore, the vibration system will automatically stop and then start again during a change in direction so that the machine does not vibrate while in NEUTRAL. It is not necessary to press the switch on the propel handle to restart the vibration in this case, it will start automatically when the propel handle is moved out of NEUTRAL. If you would like to have the machine operate in Manual Mode, consult your Cat ® dealer. Note thatin Manual Mode, the machine will vibrate in NEUTRAL and if operated this way, may damage the mat.

There are several situations in which the vibration system will not respond to a press of the switch. The vibration system will be disabled on startup and if any one or more of the following conditions exist:

- The machine is in neutral.
- The park brake is set.
- The e-stop is activated.
- The operator is out of the seat (if seat is equipped with operator presence switch).
- The engine is set to low idle.
- There is a level 3 fault on the machine controller.

If vibration has been disabled for any of the reasons above, it will be necessary to press the button on the top of the propel handle to re-enable vibration.

FWD – Push the lever away from the operator to move the compactor forward. To cause the machine to move faster, push the lever further.

STOP – Move the lever to the center STOP position to stop the machine.

REV – Pull the lever toward the operator to move the machine in reverse. To cause the machine to move faster, pull the lever further.

Note: When traveling up a slope or when experiencing high rolling resistance, if the engine begins to lug, reduce speed to keep from stalling the machine. It may also be necessary to stop one or both drums from vibrating when operating in high load conditions.

Traction Control Switch (If Equipped) (3)

Note: Traction control switch (3) will be available only for machines with double drum configuration.

Traction control switch (3) controls the traction control system. If the front drum or the rear drum begins to slip, depress traction control switch (3) with your foot to limit the slipping of the drum. The traction control system is a hydraulic system which does not manage the engine power or idle. Traction control will be active as long as the switch is pressed down. Releasing the switch will de-activate the traction control.

Note: When you operate the traction control system, reduce the propel speed until the drum stops spinning. Excessive movement of the propel lever may cause the engine to lug when the traction control system is engaged.

Note: Operating the machine with traction control active for long periods of time may cause the engine to overheat and/or burn more fuel. Use only as necessary.

12V Power Receptacle (4)



12V Power Receptacle – 12V power receptacle (4) can be used to power automotive electrical equipment or accessories. Remove the cap before use.

Note: 12V power receptacle (4) will get powered only when engine start switch is in ON position or when the machine is on.

Left Propel Control (5) (If Equipped)



Illustration 42 g06611010

(21) On/off switch for the vibratory system

On/Off Switch for the vibratory system (21) – To turn on the vibration system, press the switch. to stop the vibration system, press the switch again.

Note: This machine is set for auto vibe from the factory. Therefore, the vibration system will automatically stop and then start again during a change in direction so that the machine does not vibrate while in NEUTRAL. It is not necessary to press the switch on the propel handle to restart the vibration in this case, it will start automatically when the propel handle is moved out of NEUTRAL. If you would like to have the machine operate in manual mode, consult your Cat ® dealer. Note thatin manual mode, the machine will vibrate in NEUTRAL and if operated this way, may damage the mat.

There are several situations in which the vibration system will not respond to a press of the switch. The vibration system will be disabled on startup and if any one or more of the following conditions exist:

- The machine is in neutral.
- The park brake is set.
- The e-stop is activated.
- The operator is out of the seat (if seat is equipped with operator presence switch).
- The engine is set to low idle.
- There is a Level 3 fault on the machine controller.

If vibration has been disabled for any of the reasons above, it will be necessary to press the button on the top of the propel handle to re-enable vibration.

Note: Full speed may not be achievable in forward or reverse when the left propel handle is used. The left propel handle is only intended for working speeds.

FWD – Push the lever away from the operator to move the compactor forward. To cause the machine to move faster, push the lever further.

STOP – Move the lever to the center STOP position to stop the machine.

REV – Pull the lever toward the operator to move the machine in reverse. To cause the machine to move faster, pull the lever further.

Note: When traveling up a slope or when experiencing high rolling resistance, if the engine begins to lug, reduce speed to keep from stalling the machine. It may also be necessary to stop one or both drums from vibrating when operating in high load conditions.

Emergency Stop Knob (6)

NOTICE
Perform a walk around inspection after actuation of a shutdown device.

Take necessary corrective action to resolve the cause of the shutdown.

Ensure that no additional damage has been done or could occur before returning to operation.



Emergency Stop Knob – To stop the engine, push the knob downward. To release emergency stop knob (6) , pull the knob upward.

Work Lights (7)



Standard Light – Standard light is three position, center is OFF push the top to get the front and rear lights, push the bottom to get rear light only.

Lighting option - Roading – Optional switch is two position switch. press top to get rear work light, press bottom to turn off.

Front/Position Lights (If Equipped) (8)

Front/Position Lights – Press the switch to the left to turn on the front lights. Press the switch to the right to turn on the position lights. The middle position turns off both front and position lights.

Turn Signals (If Equipped) (9)

Turn Signals – Push down on the left side of the switch to activate the left turn signal (9). Push down on the right side of the switch to activate the right turn signal (9).

Hazard Lights (If Equipped) (10)

Hazard Lights – Push down on the right side of the switch to turn on hazard lights (10). Push down on the left side of the switch to turn off hazard lights (10).

Gauges and Indicators (11)

Refer to Operation and Maintenance Manual, Monitoring Systemfor more information.

Parking Brake (12)

Parking Brake Apply – Push down on the top side of the switch to apply parking brake (12).

Parking Brake Release – Push down on the bottom side of the switch to release parking brake (12).

Note: If the propel control is not in the NEUTRAL position when parking brake (12) is released, the machine will not propel. The propel control must be returned to the NEUTRAL position and the park brake switch must be turned back on before it may be switched off to propel the machine.

Horn (13)

Horn – Depress horn (13) button to sound. Use horn (13) for alerting personnel or for Signaling personnel.

Water Spray System (14)

NOTICE
Always check the water level in the water tanks before you use the water spray system (14).

Contaminated water in the water tanks can cause performance problems and reliability problems in the water spray system (14).

NOTICE
Drain the water system completely when freezing conditions exist in order to avoid damage to the water system components.

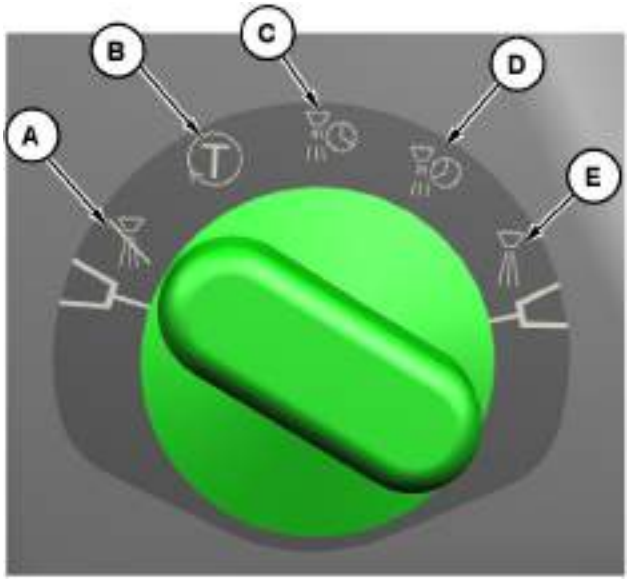


Illustration 43 g06673743

(A) Off
(B) Test
(C) Intermittent low
(D) Intermittent high
(E) Continuous on

- Move the switch to off (A) position to deactivate the spray system.

- Move the switch to test (B) position. Test (B) position is used by the operator to check the operation of the spray system. The spray system will work only if propel is in NEUTRAL position and operator out of seat and when the park brake is applied. This operation is carried by the operator to adjust the nozzles to provide proper coverage on the drum.
- Move the switch to intermittent low (C) position. Intermittent low (C) position will spray the water for 10 seconds and get turned off for the next 20 seconds. Intermittent low (C) will work only when operator is in seat and propel not in (Neutral) position and park brake released.
- Move the switch to intermittent high (D) position. Intermittent high (D) position will spray the water for 20 seconds and get turned off for the next 10 seconds. Intermittent high (D) will work only when operator is in seat and propel not in (Neutral) position and park brake released.
- Move the switch to continuous on (E) position. The spray system will work continuously when operator is in seat and propel not in (Neutral) position and park brake released.

Engine Start Switch (15)

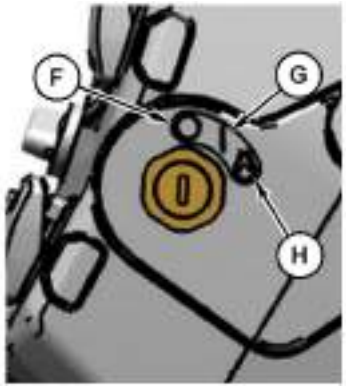


Illustration 44 g06611011

(F) Off position
(G) On position
(H) Start position

OFF – To disconnect the electrical power to the engine and to the machine, turn the switch in a counterclockwise direction to OFF position (F). Turn the switch to OFF position (F) before trying to restart the engine. Turn the switch to OFF position (F) to stop the engine.

ON – To activate the machine circuits, turn the switch to ON position (G). When the switch is released from START position (H), the switch will return to ON position (G).

START – Turn the engine start switch to START position (H) to crank the engine. Release the key when the engine starts.

Note: If the engine does not start, return the switch to the OFF position (F) before returning to the START position (G).

Refer to the Operation and Maintenance Manual, Engine Starting for more information.

Vibratory Mode (CB2.5 GCand CB2.7 GC) / Emulsion Spray (CC2.7 GC Only) (16)

Vibratory mode – The vibratory mode (16) switch is a three position switch. In the center position, both drums will vibrate when the system is activated. Push the top of the switch to vibrate the front drum only, press the bottom of the switch to vibrate only the rear drum.

Emulsion spray – The emulsion spray (16) is a momentary switch, the pump will run when the switch is pressed and the pump will stop when switch is released. Push down the top of the switch to turn the additive spray system.

Throttle Control (17)

Press the switch to the right for low idle, machine will run at 1100 RPM. Press the switch to the left for high idle, machine will run at 2800 RPM. Press the switch to mid-idle position, machine will run at 2300 RPM.

Beacon (If Equipped) (18)

Press the top of the switch to turn on beacon (18). Press the bottom of the switch to turn off beacon (18).

Digital Display (19)

Digital display (19) will display the hour meter of the machine. The display also shows that the fault code occurred in the machine and the corresponding event will be highlighted. If more than one fault occurs, the display will show the codes and hour meter in a cycle. Refer to Operation and Maintenance Manual, Monitoring Systemfor more information.

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Battery Disconnect Switch
(If Equipped)

SMCS Code: 1411

NOTICE
Never move the battery disconnect switch to the OFF position while the engine is operating. Serious damage to the electrical system could result.

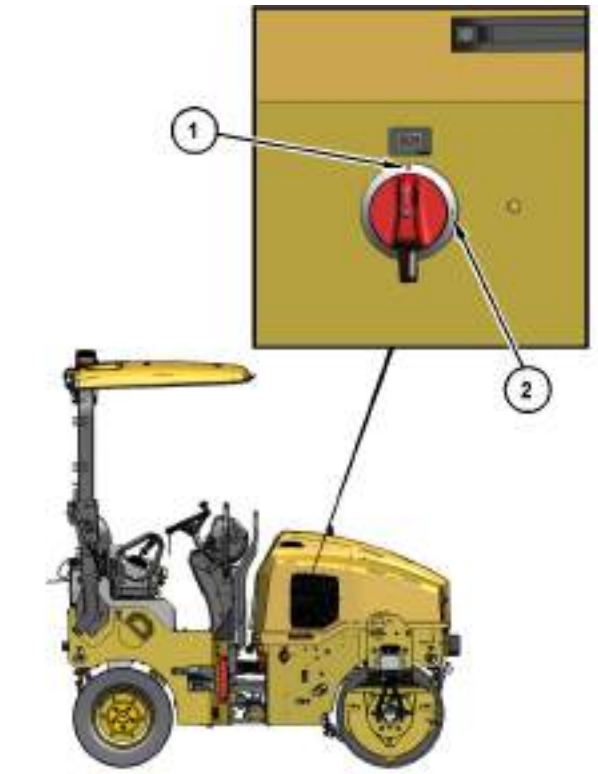


Illustration 45 g06673999

(1) Off position
(2) On position

Battery Disconnect Switch – The battery disconnect switch is on the front right side of the engine compartment.

OFF – To deactivate the electrical system, turn the switch to the OFF position (1).

ON – To activate the electrical system, turn the switch in a clockwise direction. Depending on the type of disconnect switch on the machine, it may be necessary to insert the key first. The switch must be in the ON position (2) to start the engine. Refer to “Engine Starting“ for more information.

The functions of the battery disconnect switch and the engine start switch are different. When the battery disconnect switch is turned to the OFF position (1), the entire electrical system is disabled. When the engine start switch is turned to the OFF position, the battery remains connected to the electrical system.

If the machine is equipped with a removable battery disconnect switch key, remove the key when you exit the machine for an extended time. Also, remove the key when you service the electrical system.

To ensure that no damage to the engine occurs, verify that the engine is fully operational before cranking the engine. Do not crank an engine that is not fully operational.

Perform the following procedure to check the battery disconnect switch for proper operation:

1. When the battery disconnect switch in the ON position (2).

a. Make sure that the electrical components in the operator compartment are functioning.

b. Verify that the hour meter is displaying information

c. Verify that the engine will crank.

2. When the battery disconnect switch in the OFF position (1).

a. Make sure that the electrical components are not functioning.

b. Verify that the hour meter is not displaying information.

c. Verify that engine will not crank.

consult your Cat ® dealer, If any of these functions continue to work with the battery disconnect switch in OFF position (1)

i08466561

Monitoring System

SMCS Code: 1900; 5258; 7400; 7402; 7450; 7451; 7490

The monitoring system informs the status of the machine systems, problems, or an impending problem.

Functional Test

When the engine start switch key is turned on, the monitoring system will conduct a self-test. The self-test consists of the following characteristics: Full sweep of the gauge needles from the minimum position to the maximum position, activation of all indicators and activation of the audible alarm for Level 3 warnings for one second. The operator must observe the self-test to verify that the monitoring system is working properly.

The self-testing feature verifies that the modules of the monitoring system are properly operating.

The operator must observe the outputs to determine if the modules are operating properly. The self-testing feature is 3 seconds long.

WARNING

If the action alarm does not sound during this test or machine monitoring displays are not functioning, do not operate the machine until the cause has been corrected. Machine operation with faulty action alarms or displays could result in injury or death as any Warning Category 3 notifications will not be relayed to the operator.

During the self-test, all status indicators are illuminated.

- The action light stays illuminated.
- The action alarm sounds once.

The monitoring panel is then in the normal operating mode.

Gauges and Indicators

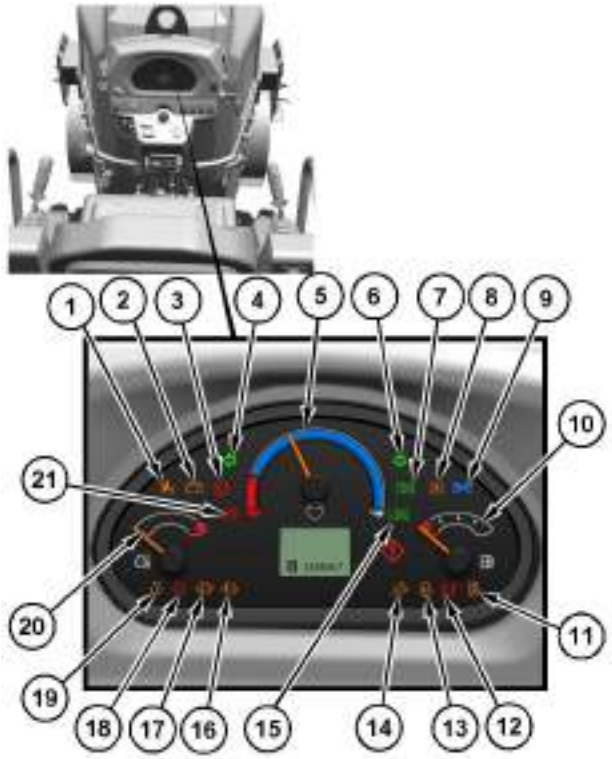


Illustration 46 g06673817

View of operator console from the operator seat

Operator Out of Seat (1) – This status indicator is illuminated when the engine is running and the operator leaves the seat unoccupied.



Check Battery (2) – This alert indicator illuminates if there is a malfunction in the electrical charging system. If this alert indicator illuminates, the system voltage is too low for normal machine operation.



Emergency Stop (3) – This indicator is illuminated when the emergency stop button is activated.

NOTICE

Perform a walk around inspection after actuation of a shutdown device.

Take necessary corrective action to resolve the cause of the shutdown.

Ensure that no additional damage has been done or could occur before returning to operation.



Left Turn (4) – This indicator starts blinking when the left turn indicator is active.



Water Spray Tank Level (5) – The water spray tank level gauge Indicates the amount of water in the water spray tank when the machine is on a leveled surface.



Right Turn (6) – This indicator starts blinking when the right turn indicator is active.



High Frequency (7) – This indicator is illuminated when vibration frequency is at maximum.



Decoupling (8) – This indicator is activated when the drum is decoupling from the ground.



Position Lights ON (9) – This indicator indicates the status of the work lights on the machine.



Fuel Level (10) – The fuel level gauge indicates the amount of fuel that remains in the fuel tank. A gauge needle in the red zone indicates a low fuel level.



Water in Fuel (11) – This alert indicator indicates a plugged fuel/water separator. Stop the engine and investigate the cause of the fault.



Hydraulic Oil Temperature High (12) – This indicator is illuminated when the hydraulic oil reaches the maximum allowed range. Operation of the hydraulic functions must be reduced to allow the hydraulic fluid to cool.



Glow Plugs Wait to Start (13) – This indicator will illuminate when the glow plugs are active.



Engine Oil Pressure Low (14) – This alert indicator will illuminate when the engine oil pressure is low. An audible alarm will sound along with the indicator. If this alert indicator illuminates, stop the machine immediately. Stop the engine, engage the parking brake, and investigate the cause.



Low Frequency (15) – This indicator is illuminated when vibration frequency is at minimum.



Emission System Malfunction (16) – This indicator will illuminate for a fault resulting from poor diesel exhaust fluid (DEF) quality, selective catalytic reduction (SCR) system tampering, SCR system fault, or an impeded exhaust gas recirculation (EGR) valve.



Diesel Particulate Filter (17) – This indicator illuminates when the amount of soot that is collected in the diesel particulate filter (DPF) is above 90%.



Engine Coolant Temperature High (18) – This indicator will illuminate when the engine coolant temperature is too high. Stop the machine at a convenient location and investigate the cause.



Check Engine (19) – This indicator is active when there is an engine malfunction. Stop the engine, engage the parking brake, and investigate the cause.



Engine Coolant Temperature (20) – The gauge needle entering the red zone indicates that the coolant temperature is too high. Increased temperatures will sound the warning alarm. Continued operation of the machine during the sounding of the warning alarm or the gauge needle in the red zone may damage the engine. Stop the machine in a safe place and investigate the cause.



Park Brake Set (21) – This status indicator indicates that the parking brake is engaged. If this indicator is illuminated, disengage the parking brake before attempting to move the machine.



Illustration 47 g06674910

Fault Indicator (22) – This indicator is illuminated when there is a malfunction in the machine system.

Liquid Crystal Display (LCD) Screen (23) – The LCD screen will display the hour meter and fault codes.

i06263982

Operation Information

SMCS Code: 1000; 7000

Operate the machine only when the operator is correctly seated in the operator compartment.

For machines that are equipped with a ROPS, a seat belt should be worn at all times during machine operation to prevent serious injury or death in the event of an accident or machine overturn. Failure to wear a seat belt during machine operation may result in serious injury or death.

For machines that are NOT equipped with a ROPS, falling from a machine could cause serious injury or death. Seat belts should be worn when driving the machine on public roads.

Operate the controls only when the engine is operating.

Inspect the machine for proper operation of controls.

Operate the machine slowly in an open area in order to inspect the protective devices.

The work area must be clear. The operator can then proceed to operate the machine.

Only the operator is allowed on the machine.

Report any needed repairs.

Maintain a safe distance from the edge of cliffs, overhangs, and slide areas.

i04984890

Backup Alarm

SMCS Code: 7406

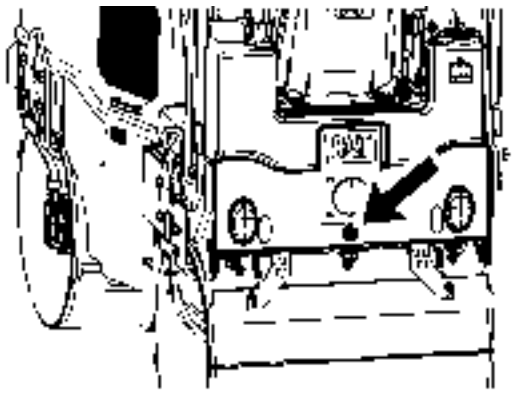


Illustration 48 g03164896

The backup alarm is located on the rear of the machine.

Backup Alarm – The backup alarm will sound when the propel lever is in the REVERSE position. The backup alarm alerts any personnel that the machine is backing up.

i08286524

Rollover Protective Structure (ROPS)
(Foldable)

SMCS Code: 7323

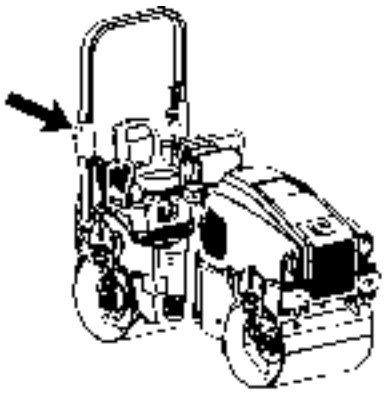


Illustration 49 g03165497

If the machine is equipped with a foldable rollover protective structure (ROPS), the ROPS must be in the raised position before you operate the machine. Before the machine is shipped, lower the ROPS in order to provide more clearance during transportation. The ROPS can be folded toward the rear of the machine.

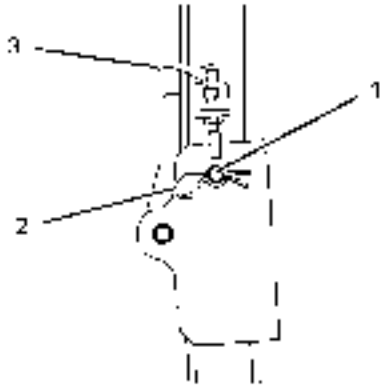


Illustration 50 g03165498

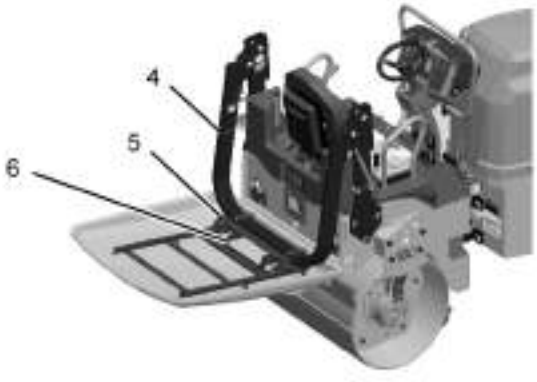


Illustration 51 g03485338

Raise

In order to raise the ROPS, perform the following steps for both folding joints on the ROPS:

1. If the machine is equipped with a canopy, perform steps 1a through 1d prior to raising the ROPS.

a. Remove the canopy pins from position (6).

b. Lower the canopy.

c. Insert the canopy pins into position (5).

d. Disconnect hold down cables (4).

2. Raise the ROPS to the full upright position. Use two people to raise the ROPS.

Note: If the ROPS is equipped with a canopy, there should be a gas springs attached to either side of the ROPS to assist with lifting and lowering the ROPS. Do not attempt to raise or lower the ROPS if the gas springs are not in place.

3. Install the locking pin (1).

4. Install cotter pin (2) through locking pin (1). The cotter pin will lock the locking pins in place.